



Commercial Escrow Agent Tools Suite

(<https://ceatools.streamlit.app/>)

Proof-of-Concept (PoC) Digital Workspace for Real Estate Transactions

Welcome to the **Commercial Escrow Agent Tools Suite**—a custom desktop dashboard engine built using Python and Streamlit. This application is designed specifically to optimize efficiency, lower transactional compliance risk, and eliminate manual bottlenecks in high-liability commercial real estate closings.



Management Executive Brief (Value Abstract)

For decision-makers and commercial escrow managers, a high-volume commercial transaction pipelines represent significant legal and financial liability. A single missed wire cutoff, unverified entity signer, or miscalculated per-diem tax proration can cause multi-million dollar deals to fall through or result in post-closing escrow claims.

This operational suite solves those vulnerabilities by automating repetitive file-auditing and time-sensitive calculations:

- **Risk Mitigation:** Scans incoming documents for critical title insurance markers and flags missing materials before closing day.
- **Default Protection:** Provides real-time countdown alerts for local and regional bank wire deadlines to ensure same-day funding settlement.
- **Operational Velocity:** Replaces spreadsheet formulas with an instant, interactive ledger that dynamically shifts settlement metrics as closing parameters evolve.



Operational Modules & User Guide

Navigate between these specialized tools using the clean sidebar layout on the left side of the screen.



1. The Welcome Workspace (Main)

- **Purpose:** Serving as the central dashboard hub, this screen guides operators into their active transaction workflows.
- **How to Use:** Review the core tool functions listed on the central panel and click any utility in the left navigation sidebar to launch a task workspace.

2. 'Clean Title' Document Tracker

- **Purpose:** Eliminates manual text-skimming by running automated phrase matching against incoming PDF files to verify closing readiness.
- **How to Use:**
 1. Drag and drop single or multiple closing documents into the **Document Upload Center** on the left sidebar.
 2. The engine instantly extracts raw text structures and checks them against the **Title Compliance Checklist** (e.g., verifying *Payoff Statements*, *Lien Releases*, *Entity Corporate Resolutions*, and *Title Insurance Commitments*).
 3. View the live **File Completion Rate** progress bar.
 4. Review **Smart Risk Alerts**—such as system warnings to verify “Good Through” dates on detected payoffs—to address issues before files proceed to funding.

3. Commercial Closing Statement Calculator

- **Purpose:** A digital sandtable to model calculations, test ALTA/HUD statement inputs, and distribute closing line-items dynamically.
- **How to Use:**
 1. Enter transaction figures under **Transaction Details** (Purchase Price, Earnest Deposits, Loan Amounts, and Property Taxes).
 2. Enter fees under **Title, Escrow & Recording Fees** and use the interactive dropdown selectors to allocate costs (*Split 50/50*, *Buyer Pays Entirely*, *Seller Pays Entirely*).
 3. Input custom entries (such as broker commissions or survey fees) under the **Dynamic Miscellaneous Adjustments** panel.
 4. Click  **Run Balancing Math** to view a perfectly formatted, double-entry **Trial Balance Ledger** tracking specific net wire requirements for buyers and sellers.

4. Critical Date & Wire Deadline Alerts

- **Purpose:** Prevents title defaults and funding delays by tracking contract milestones alongside real-time commercial banking deadlines.
- **How to Use:**
 1. Use the **Wire Deadline Settings** panel on the left sidebar to select your active regional time zone and target banking entity cutoff time.
 2. The system calculates a real-time countdown banner: **Green** for stable windows, **Yellow** for approaching limits, and **Red** for critical wire crunches or closed banking hours.
 3. Use the **Active Transaction Milestone Audit** form to input contract dates and window durations. The calculation engine will map exact calendar deadlines for title objections, financing contingencies, and closing days.

5. Entity Verification & KYC Assistant

- **Purpose:** Organizes and verifies signature paths for complex signing structures (such as multi-tier LLCs, corporate managing members, and trusts).
- **How to Use:**
 1. Fill out the **Add Signing Authority Record** form with the entity name, authorized individual signer, and exact title capacity.
 2. Check off the specific organizational documents reviewed during your compliance check.
 3. Click **+ Log Signer Authority** to save the record down into an interactive, digital **Transaction Signing Authority Registry** log.
 4. Use the built-in **Underwriting Verification Reminders** at the bottom of the page to maintain full regulatory compliance for LLC and corporate files.

Technical Infrastructure & Local Installation

This application runs locally or via your preferred cloud hosting provider, requiring minimal resource overhead.

Prerequisites

Ensure you have **Python 3.9+** installed on your system.

Installation Steps

1. Clone the Repository:

```
git clone https://github.com  
cd Commercial-Escrow-Agent-Suite
```

bash

2. Install Required Package Dependencies:

```
pip install -r requirements.txt
```

bash

3. Launch the Local Core Server:

```
streamlit run Main.py
```

bash

The application will boot up immediately and display inside a new window in your default web browser.

Technology Stack

- **UI Framework:** [Streamlit](#) (Data-focused reactive dashboard architecture)
- **PDF Extraction Processing Engine:** [pdfplumber](#) (High-accuracy character and positional text parsing)
- **Timezone & Clock Management:** [pytz](#) (Precise Olson-database timezone mapping)
- **Data Structures:** [Pandas](#) (In-memory structured data tables)

Screenshots

Main

Document Tracker
Closing Calc
Deadline Alerts
Entity Verifier



Commercial Escrow Operations Suite

Proof of Concept Workflow Productivity Tools

Welcome to your Digital Escrow Assistant

Select a tool from the sidebar navigation to streamline your daily commercial closing pipeline:

- 🔍 **Document Tracker:** Scan PDF files to automatically verify required title documents and look for expiration dates.
- 📊 **Closing Calculator:** Simulate closing dates, calculate interest prorations, and audit debit/credit balances.
- 🕒 **Deadline Alerts:** View a daily countdown of wire cutoff times and critical contract milestones.
- 📄 **Entity Verifier:** Map out complex corporate signing structures and cross-reference authorization documents.

💡 **Tip:** Use the sidebar menu on the left to seamlessly jump between different active transaction tasks.

Main

Document Tracker
Closing Calc
Deadline Alerts
Entity Verifier

Document Upload Center

Drag and drop closing PDFs here



Main

Document Tracker
Closing Calc
Deadline Alerts
Entity Verifier

'Clean Title' Document Tracker

Successfully loaded 1 document(s). Processing text extraction...



Title Compliance Checklist

System audit based on automated phrase matching:



Payoff Statement

Matches found in the uploaded text pool.



Lien Release / Satisfaction

Matches found in the uploaded text pool.



Entity Authorization (LLC/Corp)

Matches found in the uploaded text pool.



Title Insurance Commitment



Audit Summary & Flagged Risks

File Completion Rate

100%



Smart Risk Alerts

Payoff Expiration Risk: A payoff statement was detected. Ensure the 'Good Through' date covers the projected wire date.



Commercial Closing Statement Calculator



Transaction Details

Purchase Price (\$)

5000000

Target Closing Date

2026/06/17

Earnest Money Deposit (\$)

250000

Annual Property Tax (\$)

24000

New Lender Loan Amount (\$)

3500000

Lender Per Diem Interest Rate (%)

6.500

Days of Prepaid Interest to Collect

15



Title, Escrow & Recording Fees

Title Insurance Premium (\$)

12500

Title Endorsement Charges (\$)

1500

Who pays Title Premium?

Split 50/50

Who pays Endorsements?

Buyer Pays Entirely

Kensington Vanguard Settlement/Escrow Fee (\$)

Government Recording Fees (\$)

Main

Document Tracker

Closing Calc

Deadline Alerts

Entity Verifier

Wire Deadline Settings

Your Operating Timezone

Eastern Time (NY Corp HQ)

Bank Processing Cutoff

Commercial Settlement Bank (T...

Critical Date & Wire Deadline Alerts

Real-Time Wire Deadline Tracking

Your Local Desk Time

05:44 PM (EDT)

Selected Target Deadline

04:30 PM (EDT)

Same-Day Processing Closed: The 04:30 PM deadline has passed. Wires moved now will default to next business day credit.

Active Transaction Milestone Audit

Input deal milestones to instantly calculate operational safe windows and target timelines.

Transaction Name / File Number	Due Diligence Window (Days)
File #2026-884A (Commercial Plaza)	30
Contract Effective Date	Financing Contingency Window (Days)
2026/06/03	45
Scheduled Closing Date	Title Objection Review Window (Days)
2026/07/18	10

Compute Timeline Map

Entity Verification & KYC Assistant

Escrow Protocol: Use this module to map out multi-layer entity structures (e.g., a subsidiary LLC managed by a parent Corp) and log which resolutions, operating agreements, or incumbency certificates have been verified for closing day.

Add Signing Authority Record

Name of Contracting Entity	Parent Entity / Managing Member (if Applicable)
e.g., Acme Commercial Holdings, LLC	e.g., Acme Capital Corp (Parent)
Authorized Individual Signer Name	Corporate Documents Reviewed & Verified
e.g., John Doe	Choose options
Corporate Title / Capacity	Authority Verification Status
e.g., Managing Member / Vice President	Verified - Fully Authorized

+ Log Signer Authority

Transaction Signing Authority Registry

No signing structures logged yet. Use the form above to build the escrow authority map.

Main

Document Tracker

Closing Calc

Deadline Alerts

Entity Verifier

Disclaimer: This software is provided as an operational Proof of Concept (PoC) tool designed to assist escrow professionals with pipeline data analysis. It should be used alongside primary internal title underwriting platforms.